LumiLeaf Production Deployment Handbook

1. Purpose

Guide for deploying LumiLeaf from a developer laptop to a production environment used by factory staff.

2. Current Development Workflow

Developer PC: IntelliJ IDEA + XAMPP (Apache/MySQL) + Spring Boot. Access via localhost:8080. Suitable only for development.

3. Production Architecture

Employees -> HTTPS Domain -> Nginx -> Spring Boot Application -> MySQL Database. Application runs as a Linux service; no IntelliJ or XAMPP.

4. Infrastructure Checklist

Ubuntu Server 24.04 LTS, Java 21, MySQL 8/MariaDB, Nginx, SSL certificate, domain, UPS, automated backups, firewall, SSH access.

5. Folder Structure

/opt/lumileaf/app (JAR)
/opt/lumileaf/uploads (reports, QR, images)
/opt/lumileaf/config
/opt/lumileaf/logs
/opt/lumileaf/backups

6. Database

Separate MySQL server/database. Never package data inside the application. Backup daily and test restores.

7. First Deployment

Build with Maven, upload JAR, configure application-prod.properties, start via systemd, configure Nginx and HTTPS, verify application.

8. Updating Production

Develop locally, commit to Git, test in staging, backup DB, replace JAR, restart service, run Flyway/Liquibase migrations if schema changes, verify.

9. Zero-Downtime

Use blue-green or rolling deployments when continuous availability is required. Small internal systems may accept a short restart window.

10. Rollback

Keep previous JAR, backup database before migration, revert to previous version if validation fails.

11. Security

HTTPS, strong passwords, RBAC, audit logs, regular OS updates, least-privilege DB accounts.

12. Monitoring

Monitor CPU, RAM, disk, application logs, database health, backups, SSL expiry.

13. CI/CD

GitHub -> Build -> Automated tests -> Package JAR -> Deploy to staging -> Approve -> Production.

14. Operational Workflow

Developer -> Git -> Build -> Test -> Backup -> Deploy -> Migrate -> Verify -> Users continue working.

15. Future Improvements

Docker, Kubernetes, centralized logging, Prometheus/Grafana, high availability database, object storage for uploads.